

Q) Define null space and find spanning set for the null space of given matrix.

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Ans) "Null Space"

• null space of an $m \times n$ matrix A is the set of all solution to the homogeneous equation, $Ax=0$

it is represented in set notation form as,

$$\text{Nul } A = \{x : x \text{ is } \mathbb{R}^n \text{ and } Ax=0\}$$

Problem

Given matrix is,

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

The augmented matrix $[A \ 0]$ is represented as,

$$[A \ 0] = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 & 0 \\ 1 & -2 & 2 & 3 & -1 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{bmatrix}$$

$$R_2 \rightarrow 3R_2 + R_1, R_3 \rightarrow 3R_3 + 2R_1,$$

$$= \begin{bmatrix} -3 & 6 & -1 & 1 & -7 & 0 \\ 0 & 0 & 5 & 10 & -10 & 0 \\ 0 & 0 & 13 & 26 & -26 & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2/5, R_3 \rightarrow R_3/3$$

$$= \begin{bmatrix} -3 & 6 & -1 & 1 & -7 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$= \begin{bmatrix} -3 & 6 & -1 & 1 & -7 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1/3$$

$$= \begin{bmatrix} 1 & -2 & 1/3 & -1/3 & 7/3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 1/3 R_2$$

$$= \begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The equations are,

$$x_1 - 2x_2 - x_4 + 3x_5 = 0$$

$$x_1 = 2x_2 + x_4 - 3x_5$$

$$x_3 + 2x_4 - 2x_5 = 0$$

$$\Rightarrow x_3 = -2x_4 + 2x_5$$

$\Rightarrow$ And x_2, x_4, x_5 are free variables.

$\Rightarrow$ The general solution can be written as linear combination of vectors with free variables.

$$\Rightarrow x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix}$$

$$\Rightarrow x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $u \quad \quad v \quad \quad w$

$$\Rightarrow x = x_2 u + x_4 v + x_5 w$$

Since every linear of u, v, w is an element of $\text{null } A$

$\therefore \{u, v, w\}$ is spanning set for $\text{null } A$.

2) State and Prove basis theorem.

Statement: if V is a P -dimensional vector space such that $P \geq 1$ then,

i) any linearly dependent set of exactly P elements in V is automatically a basis for V and.

ii) any set of exactly P element that spans V is automatically a basis for V .

Proof

Given that

V is a P -dimensional vector space i.e. $\dim V = P$ and $P \geq 1$

i) let S be the linearly independent set of vector (P -element)

Case i) if S spans V

$\Rightarrow S$ is a basis for V .

Case ii) if S does not spans V

then,

a linearly independent set S of P elements can be extended to a basis for V

i.e. extended basis of V consist more than P element

which is a contradiction to $\dim V = P$

then,

a) Since, by Kuro theorem, any linearly independent set in subspace H can be extended to a basis for H and $\dim(H) \leq \dim(V)$, but the number of vectors in expanded linearly independent set can never exceed the dimension of V .

Also, $\dim V = p$ and basis must contain exactly p elements.

$\therefore S$ is automatically a basis for V .

ii) Let S has p elements

Since, $V \neq \{0\}$

From the spanning set theorem

If S' is a linearly independent set then the subset S' of S is a basis for V .

If S' is linearly dependent then set formed from S by removing v_k , which is a linear combination of remaining vectors then S' is a basis for V which is a contradiction.

$\Rightarrow \dim V = p$

$\Rightarrow$ subset S' must contain p elements

$\therefore S = S'$

$\therefore$ Any set of exactly p elements that span V is automatically a basis for V .

5) State and Prove Spanning set theorem.

5A) Let $S = \{v_1, v_2, \dots, v_p\}$ be a set in V & $H = \text{Span} \{v_1, v_2, \dots, v_p\}$

i) if one of the vectors in S i.e. v_k is a linear combination of the remaining vectors in S , then the set formed from S by removing v_k still spans H .

ii) If $H \neq \{0\}$, then some subset of S is a basis for H .

PROOF

Given that

$S = \{v_1, v_2, \dots, v_p\}$ is a set in V

where, $v_1, v_2, \dots, v_p$ - vectors in S

and $H = \text{Span}(v_1, v_2, \dots, v_p)$

if v_p is the linear combination of $v_1, v_2, \dots, v_{p-1}$ then

$$v_p = a_1 v_1 + a_2 v_2 + \dots + a_{p-1} v_{p-1} \quad \text{--- scalar ---} \rightarrow (1)$$

where, $a_1, a_2, \dots, a_{p-1}$

consider an arbitrary element x in H such that:

$$x = c_1 v_1 + c_2 v_2 + \dots + c_{p-1} v_{p-1} + c_p v_p \rightarrow (2)$$

where

$c_1, c_2, \dots, c_p$ scalar

from eq (2)

$$\begin{aligned}
 x &= C_1 V_1 + C_2 V_2 + \dots + C_{p-1} V_{p-1} + C_p (a_1 V_1 + a_2 V_2 + \dots + a_{p-1} V_{p-1}) \\
 &= (C_1 + a_1 C_p) V_1 + (C_2 + a_2 C_p) V_2 + \dots + (C_{p-1} + a_{p-1} C_p) V_{p-1}
 \end{aligned}$$

Clearly 'x' is a linear combination of $V_1, V_2, \dots, V_{p-1}$

Thus, $\{V_1, V_2, \dots, V_{p-1}\}$ still spans H .

ii) Consider the original spanning set S as linearly independent then it consists a basis for H .

Two or more vectors in the spanning set can repeat the process until it's linearly independent. Thus, the basis of S gets reduced to one-zero vector. This due to existence of span vectors in H .

$$\text{i.e. } H \neq \{0\}$$

→ Show that Range of T is a subspace of W .

Ans) Given that

$T: V \rightarrow W$ is a linear transformation.

Range of T is given as,

$$\text{Range } T = \{T(x) / x \in V\} \rightarrow 0$$

Range of T is a subspace of W if it satisfies the following three properties.

i) 0 is in range T .

ii) $x+y$ is in range T .

iii) αx is in range T .

i) Since, T is a linear transformation,

$$T(0_V) = 0_W$$

$\Rightarrow 0_W \in \text{Range } T$

Let $T(x), T(y) \in \text{Range } T, x, y \in V$

consider,

$$T(x+y) = T(x) + T(y)$$

$$= x + y$$

$$\in V$$

$\therefore T(x+y) \in \text{Range } T$

iii) Let c be any scalar

consider,

$$T(cx) = cT(x)$$

Since T is linear transformation

and $x \in V$

$$\Rightarrow cx \in V$$

$\therefore T(cx) \in \text{Range } T$

$\therefore \text{Range } T$ is a subspace of W

10) Let $B = (b_1, b_2, \dots, b_n)$ be a basis for vector space V . then prove that for each $x \in V$, there exist a unique set of scalar $c_1, c_2, \dots, c_n$ such that $x = c_1 b_1 + c_2 b_2 + \dots + c_n b_n$.

10a) Given that

V is a vector space

and $B = (b_1, b_2, \dots, b_n)$ is a basis for V

i.e. B is linearly independent & span V

then, for each $x \in V$, there exist scalar $c_1, c_2, \dots, c_n$ such that

$$x = c_1 b_1 + c_2 b_2 + \dots + c_n b_n \rightarrow (1)$$

Let, there exist scalar $d_1, d_2, \dots, d_n$ such that

$$x = d_1 b_1 + d_2 b_2 + \dots + d_n b_n \rightarrow (2)$$

From eq (1) & (2)

$$c_1 b_1 + c_2 b_2 + \dots + c_n b_n = d_1 b_1 + d_2 b_2 + \dots + d_n b_n$$

$$c_1 b_1 + c_2 b_2 + \dots + c_n b_n - (d_1 b_1 + d_2 b_2 + \dots + d_n b_n) = 0$$

$$(c_1 - d_1) b_1 + (c_2 - d_2) b_2 + \dots + (c_n - d_n) b_n = 0 \rightarrow (3)$$

Since, B is linearly independent, then all the weight in eq (3) must be zero

$$\text{i.e. } c_1 - d_1 = 0, c_2 - d_2 = 0, \dots, c_n - d_n = 0$$

$$c_1 = d_1, c_2 = d_2, \dots, c_n = d_n$$

$$c_j = d_j, 1 \leq j \leq n$$

$\therefore$ i.e. evaluation (1) & (2) have same representations

$\therefore$ For every $x \in V$ there exist some scalar $c_1, c_2, \dots, c_n$ such that $x = c_1 b_1 + c_2 b_2 + \dots + c_n b_n$.

$\therefore$ This representation is unique.

28) 28) Prove that a set $\{v_1, v_2, \dots, v_p\}$ of two or more vectors with $v_1 \neq 0$ is linearly dependent if and only if some v_j with $j > 1$ is a linear combination of the preceding vectors $v_1, v_2, \dots, v_{j-1}$.

28) Given that

Consider a set $\{v_1, v_2, \dots, v_p\}$ of two or more non-zero vectors of a vector space V .

Let the set be linearly dependent.

then, for $j > 1$, the linear combination v_j is given by,

$$a_1 v_1 + a_2 v_2 + \dots + a_{j-1} v_{j-1}$$

where $a_1, a_2, \dots, a_{j-1}$ are non-zero scalars

the vectors after v_{j-1} do not exist, as the scalars are zero i.e. for linear dependence

$$a_1 v_1 + a_2 v_2 + \dots + a_{j-1} v_{j-1} + (-1) \cdot 0 v_{j+1} + \dots + 0 v_n = 0$$

$\therefore$ The vector v_j is a linear combination $\left[\because \sum_i a_i v_i = \right]$
of the preceding vectors.